WORKED SOLUTION

STEP II 1998, Question 7

Pure • Inequalities via Derivatives

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q7

The key idea

Each inequality is proved the same way: the function is zero at $x = 0$ and its derivative is positive, so the function goes up from zero. The neat part is that the functions are chained. The derivative of g turns out to involve f, and the derivative of h turns out to be g evaluated at $2x$. So each part gives you the positive derivative you need for the next one. In part (iv), taking logarithms turns the quotient F into a sum, and the sign of F'/F is the sign of the expression from part (iii).

Solution

The tool used throughout is this: if ϕ is differentiable, $\phi(0) = 0$ and $\phi'(x) > 0$ for $0 < x < a$, then ϕ is strictly increasing on $[0, a)$, so $\phi(x) > \phi(0) = 0$ for $0 < x < a$. (By the mean value theorem, $\phi(x) = \phi(x) - \phi(0) = x\,\phi'(c)$ for some $c \in (0, x)$, and this is positive.)

Part (i)

We have $\mathrm{f}(0) = \tan 0 - 0 = 0$, and

$$\mathrm{f}'(x) = \sec^2 x - 1 = \tan^2 x.$$

For $0 < x < \frac{1}{2}\pi$, $\tan x$ is defined and positive, so $\mathrm{f}'(x) > 0$. Hence $\mathrm{f}(x) > 0$ for $0 < x < \frac{1}{2}\pi$; that is, $\tan x > x$ there.

Part (ii)

Again $\mathrm{g}(0) = 2 - 2 - 0 = 0$. Differentiating, using the product rule on $x \sin x$,

$$\begin{aligned}\mathrm{g}'(x) &= 2\sin x - (\sin x + x\cos x)\\ &= \sin x - x\cos x\\ &= \cos x\,(\tan x - x) = \cos x\,\mathrm{f}(x).\end{aligned}$$

For $0 < x < \frac{1}{2}\pi$ we have $\cos x > 0$, and $f(x) > 0$ by part (i). So $g'(x) > 0$, and since $g(0) = 0$ it follows that $g(x) > 0$ for $0 < x < \frac{1}{2}\pi$.

Part (iii)

Once more $h(0) = 0$. Differentiating,

$$\begin{aligned} h'(x) &= 2 + \cos 2x - 2x \sin 2x - 3\cos 2x \\ &= 2 - 2\cos 2x - 2x \sin 2x. \end{aligned}$$

? **Ask yourself** Have you seen this expression before?

It is g with x replaced by $2x$:

$$h'(x) = 2 - 2\cos(2x) - (2x)\sin(2x) = g(2x).$$

If $0 < x < \frac{1}{4}\pi$ then $0 < 2x < \frac{1}{2}\pi$, so $g(2x) > 0$ by part (ii). This is why the range is now $\frac{1}{4}\pi$ rather than $\frac{1}{2}\pi$. So $h'(x) > 0$ on $0 < x < \frac{1}{4}\pi$, and with $h(0) = 0$ this gives $h(x) > 0$ for $0 < x < \frac{1}{4}\pi$.

For the "hence", rewrite the given expression using double angles. Since $\sin^2 x + 3\cos^2 x = 1 + 2\cos^2 x = 2 + \cos 2x$ and $3 \sin x \cos x = \frac{3}{2}\sin 2x$,

$$\begin{aligned} x(\sin^2 x + 3\cos^2 x) - 3\sin x \cos x &= x(2 + \cos 2x) - \tfrac{3}{2}\sin 2x \\ &= 2x + x\cos 2x - \tfrac{3}{2}\sin 2x = h(x). \end{aligned}$$

So the expression is exactly $h(x)$, which is positive for $0 < x < \frac{1}{4}\pi$, as required.

Part (iv)

For $0 < x < \frac{1}{4}\pi$, the quantities x, $\sin x$ and $\cos x$ are all positive, so $F(x) > 0$ and we may take logarithms:

$$\ln F(x) = \ln x + \tfrac{1}{3}\ln\cos x - \ln\sin x.$$

Differentiating both sides with respect to x,

$$\frac{F'(x)}{F(x)} = \frac{1}{x} - \frac{\sin x}{3\cos x} - \frac{\cos x}{\sin x}.$$

? **Ask yourself** How do we see the sign of this?

Put it over the common denominator $3x \sin x \cos x$, which is positive on this interval:

$$\begin{aligned} \frac{F'(x)}{F(x)} &= \frac{3\sin x \cos x - x\sin^2 x - 3x\cos^2 x}{3x \sin x \cos x} \\ &= -\frac{x(\sin^2 x + 3\cos^2 x) - 3\sin x \cos x}{3x \sin x \cos x} = -\frac{h(x)}{3x \sin x \cos x}. \end{aligned}$$

By part (iii) the numerator $\mathrm{h}(x)$ is positive and the denominator is positive, so $\mathrm{F}'(x)/\mathrm{F}(x) < 0$. Since $\mathrm{F}(x) > 0$, multiplying through gives $\mathrm{F}'(x) < 0$ for $0 < x < \frac{1}{4}\pi$, as required.

Checks

- Spot values: $\mathrm{F}(0.1) \approx 0.999998$, $\mathrm{F}(0.5) \approx 0.99849$ and $\mathrm{F}(\frac{1}{4}\pi) \approx 0.98954$, so F is decreasing, but only slowly.
- Near 0 the inequalities are very tight. Expanding in Maclaurin series gives $\mathrm{h}(x) = \frac{4}{15}x^5 + \cdots$, and the x, x^3 terms cancel. That is why you can't simply "see" that $\mathrm{h} > 0$ and need the chain of derivatives. Numerically, $\mathrm{h}(0.01)/0.01^5 \approx 0.26666$.
- We tested all four inequalities, and the identities $\mathrm{g}' = \mathrm{f}\cos x$, $\mathrm{h}'(x) = \mathrm{g}(2x)$ and $\mathrm{F}'/\mathrm{F} = -\mathrm{h}/(3x\sin x\cos x)$, at 20 000 random points.

Common slips

- Arguing only that $\mathrm{f}'(x) \geq 0$, or forgetting to state $\mathrm{f}(0) = 0$. Both facts are needed: the zero start and the strictly positive derivative.
- Using part (ii) in part (iii) without checking that $2x$ lies in $(0, \frac{1}{2}\pi)$. That is exactly where the restriction $x < \frac{1}{4}\pi$ comes from.
- Concluding $\mathrm{F}' < 0$ from $\mathrm{F}'/\mathrm{F} < 0$ without noting that $\mathrm{F} > 0$.

Beyond STEP

The principle "zero at the start and positive derivative means positive" is a consequence of the mean value theorem, one of the first big theorems in a university analysis course. Repeating it, as here, is a standard way of proving inequalities such as $\sin x > x - \frac{1}{6}x^3$, and it is closely related to Taylor's theorem with remainder. Abbott's *Understanding Analysis* covers the mean value theorem and its uses carefully.

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